BloomSocial

Laboratory Work 7 Final Exam Project Planning and UX/UI Design

Discipline:Developmentof WebApplicationsinJavaSpring

Team: Arysbekkyzy Uasilya; Amankos Aikorkem

Work format: Team of 2 students

Institution: International Information Technology University

Date: October 2026

Figma design: https://www.figma.com/design/2SBsOY1CHzkCiOpImaEqBg GitHub

repository: https://github.com/IM-TIRED-EVERYDAY/BloomSocial-Lab7.git

1. Project Description

BloomSocial is a web-based social networking platform designed to help people connect, share personal updates, and interact within a friendly digital community. The application will provide account registration and secure login, customizable user profiles, a personalized home feed, and tools for publishing posts with optional images. Users will be able to explore posts, leave comments, and express appreciation through likes. Profile editing will let members maintain accurate personal information and choose a profile picture. The proposed visual style uses soft pink and lavender accents, clear typography, and consistent navigation to create an approachable experience. BloomSocial is particularly suitable for students and young adults who want a simple space to share interests, activities, and achievements without complicated workflows. The future system will use Java Spring Boot with a layered architecture, PostgreSQL for persistent data storage, and Thymeleaf for rendering responsive web pages. Spring Security will protect authenticated actions, while validation will improve input reliability. This laboratory work focuses on defining the solution, documenting its architecture and development plan, and producing an interactive Figma prototype rather than implementing a working backend. The resulting proposal and design will guide development and testing during the Final Exam project.

2. Objectives, Target Users and Problem

Main objectives: build a secure and easy-to-use social platform; support user-generated posts and profile management; enable likes and comments; ensure maintainable Spring architecture; provide

consistent responsive UX.

Target users: university students, young adults, interest-based communities and beginner social-network users.

Problem: users need a straightforward, welcoming environment for sharing updates and discovering content, with clear controls and reliable account management.

3. Functional Requirements and Expected Results

FR1. Register a new user with validated details.

FR2. Authenticate and log out securely.

FR3. View a feed of recent posts.

FR4. Create posts with text and optional images.

FR5. View and edit personal profiles.

FR6. View post details and add comments.

FR7. Like and unlike posts.

FR8. Restrict editing and deletion to the content owner.

FR9. Persist users, posts, comments and likes in PostgreSQL.

Expected results: a usable full-stack Spring Boot application for the final examination, supported by a consistent UI, database schema, tested business rules, and a demonstration-ready user journey.

4. UX/UI Design and User Flow

Figma: https://www.figma.com/design/2SBsOY1CHzkCiOpImaEqBg

Screen	Purpose
01 Login	Email/password authentication and registration link
02 Registration	Account creation with validation
03 Home Feed	Browse posts and access navigation
04 User Profile	View personal details and published posts
05 Create Post	Compose text and attach an image
06 Post Details	Read a post, likes and comments
07 Edit Profile	Update profile information

Primary clickable flow: Registration → Login → Home Feed → Create Post → Post Details; Profile → Edit Profile → Profile. The design uses native editable Figma text and frames with click-based navigation connections.

Design system: blush #FFF8FC, pastel lavender #F6E7F1, accent pink #B66BA0, dark ink #30263C, muted text #82748A; Inter typography, rounded cards, consistent spacing and buttons.

5. Technology Stack and Dependencies

Component / dependency	Choice and rationale
Programming language	Java — strongly typed, mature ecosystem
Backend	Spring Boot — configuration, embedded server
Web	Spring MVC / Spring Web — routing and controllers
Frontend	Thymeleaf, HTML, CSS, JavaScript — server-rendered UI
Database	PostgreSQL — persistent relational data
Persistence	Spring Data JPA / Hibernate — entity mapping and repositories
Build	Maven — dependencies and reproducible builds
Version control	Git / GitHub — collaboration and history
Testing	JUnit, Spring Boot Starter Test, Postman — automated and API tests
Security	Spring Security — authentication and authorization
Validation	Spring Validation — enforce form constraints
Productivity	Lombok — reduce boilerplate
Database driver	PostgreSQL Driver — JDBC connectivity

6. Layered System Architecture

Data flow: Browser / Thymeleaf UI → Spring MVC Controller → Service → Spring Data JPA Repository → PostgreSQL. Responses return from the database through Repository and Service to Controller, which renders a Thymeleaf view. Forms submit HTTP requests to MVC endpoints.

Controller: maps requests and responses, populates templates. Service: enforces business rules and permissions. Repository: performs CRUD and queries. Entity / Model: maps Java domain objects to relational tables. Database: stores persistent data.

Proposed entities: User(id, firstName, lastName, email, passwordHash, bio, profilePhoto); Post(id, authorId, content, imageUrl, createdAt); Comment(id, postId, authorId, text, createdAt); PostLike(id, postId, userId). User 1:N Post; User 1:N Comment; Post 1:N Comment; Post 1:N PostLike.

7. Team Members and Responsibilities

Member	Responsibilities
Arysbekkyzy Uasilya — Backend & Security	Spring Boot setup, MVC Controller, Service, security, backend validation; database entity/repository collaboration; integration and backend tests.
Amankos Aikorkem — Frontend & UX/UI	Figma screens and prototype, Thymeleaf templates, HTML/CSS/JS; database schema and data setup; UI integration and frontend tests.

Both students will review the database model, test core user flows, understand the complete architecture, and participate in the final presentation.

8. Development Roadmap

Stage	Work	Lead
1	Requirements and project planning	Both
2	UX/UI design and clickable prototype	Aikorkem
3	Spring Boot and Maven setup	Uasilya
4	PostgreSQL schema and JPA entities	Aikorkem + Uasilya
5	Repository, Service and Controller	Uasilya
6	Thymeleaf frontend implementation	Aikorkem
7	Integration, JUnit and manual testing	Both
8	Final presentation and demonstration	Both

9. Submission and Final Exam Preparation

Deliverables: (1) this PDF proposal, (2) editable Figma screens and prototype, (3) technology stack and dependencies, (4) architecture and data flow, (5) named team responsibilities, (6) eight-stage roadmap, and (7) GitHub repository link once created.

Current status: planning and UX/UI stage. Application code, production database, automated tests and GitHub repository are planned future deliverables; they are not represented as completed work.